

Question number	Answer	Additional guidance	Mark
5(a)(i)	Choose an item • B (increased number of slow twitch muscle fibres) is the correct answer A is not the correct answer as a decreased glomerular filtration rate is not an adaptation to run for long distances C is not the correct answer as narrower airways are not an adaptation to run for long distances D is not the correct answer as smaller cardiac output is not an adaptation to run for long distances		(1)

Question number	Answer	Additional guidance	Mark
5(a)(ii)	An explanation that includes four of the following points: • increase in (blood) carbon dioxide / decrease in (blood) pH} (1) • detected by chemoreceptors which send (more) impulses to {medulla (oblongata) / CVC} (1) • which send impulses to {SAN / intercostal / diaphragm / breathing} muscles (1) • SAN causes more {electrical impulses / depolarization / waves of excitation} (to spread over walls of atria) (1) • {cardiac muscle contraction / heart muscle	ACCEPT impulses via sympathetic NS ignore {signal / message}	(4)

	contraction / systole} occurs more frequently (1) • more frequent contraction of {intercostal muscles / diaphragm} (1)	ignore force of contraction ACCEPT {faster / more} contraction of {intercostal / diaphragm / breathing muscles} ignore force of contraction	
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Question number	Answer	Additional guidance	Mark
(b)(i)	A description that includes the following points: - as mass increases the resting metabolic rate increases (1) - as mass increases pulse rate decreases (1) - as resting metabolic rate increases pulse rate decreases (1)	ACCEPT positive correlation ACCEPT negative correlation ACCEPT negative correlation	(3)
Question number	Answer	Additional guidance	Mark
(b)(ii)	A description that includes three of the following points: - thermoreceptors detect {increase/ change} in temperature / temperature above 37°C} (1) - impulses to {thermoregulatory centre / hypothalamus / heat loss centre} (1) (thermoregulatory centre) sends impulses {via motor neurones / via sympathetic neurons / to correctly named effector} (1) - correct response (by effectors) to {increase heat loss / decrease heat energy gain} from the body (1)	ignore control centre unqualified e.g. sweat glands, hair erector muscles, liver, smooth muscles in skin blood vessels ignore signals / messages / information e.g. vasodilation, sweat production, decreased metabolic rate	(3)

